

TalentSprint

Capstone Project Report

Return Risk Analyzer

E-commerce Return & Seller Blacklist Prevention System

Identifying Misleading Product Descriptions Through NLP & Machine Learning to Reduce Returns and Protect Seller Reputation

Github repository : https://github.com/AnjaliBart/TS_Capstone_Final-Project_Anjali/tree/main

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Executive Summary

The Return Risk Analyzer is an NLP-powered system that addresses two interconnected and costly problems in online retail: misleading product descriptions that drive customer returns, and the seller blacklisting or account suspension that inevitably follows when return rates become elevated.

Two Business Problems. One Solution.

Problem 1 Customer Returns: Misleading listings create an expectation gap. When the product delivered does not match what was described, customers return it. This costs sellers 20% and 65% of an item's original price per return in logistics and restocking fees.

Problem 2 Seller Blacklisting: E-commerce platforms (Amazon, Flipkart, Myntra) automatically monitor seller return rates. Sellers who exceed platform thresholds (typically 8-15%) face listing removal, account suspension, or permanent blacklisting. This can destroy a seller's entire business overnight. The Return Risk Analyzer detects high-risk listings before customers are harmed, protecting both buyer satisfaction and seller reputation simultaneously.

The system analyzes a product description and its customer reviews to: extract and verify specific product claims, predict the likely return reasons using a supervised ML classifier, generate a 0-100 return risk score, and output actionable description fixes for the seller.

Dataset: Amazon US Customer Reviews Apparel and Beauty categories (~120,000 reviews, Kaggle).
Model: TF-IDF + Logistic Regression, macro F1 ~0.71 across 6 return reason classes. **Deployment:** Streamlit Cloud.

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1. Problem Statement & Business Context

1.1 The Return Epidemic in E-commerce

Product returns represent one of the most significant operational and financial burdens in online retail. Unlike physical stores, online shoppers cannot touch, try, or inspect products before purchase. They rely entirely on product descriptions, images, and reviews. When those descriptions are inaccurate, exaggerated, or incomplete, the gap between expectation and reality drives returns at scale.

Metric	Statistic	Source
Avg. return rate (India)	25-40%	E-commerce industry, 2024
Avg. logistics cost / return	₹150-400	Logistics + restocking
Expectation mismatch returns	~35% of returns	Industry survey
Size / fit returns (Apparel)	~30% of returns	Category analysis
Beauty returns (misdescribed)	~24% of returns	Category analysis

1.2 The Seller Blacklisting Problem

High return rates do not just hurt sellers financially through logistics costs they trigger automated platform enforcement mechanisms that can permanently end a seller’s business. This is the second, and often more severe, consequence of misleading product descriptions.

The cruel irony is that many sellers are unaware their descriptions are misleading. They copy manufacturer specifications, use stock photography, or write aspirational marketing copy without realising that customer experience will contradict these claims. By the time return rate metrics flag the problem, the damage to their seller metrics is already done.

1.3 Why Existing Solutions Fall Short

Current platform tools provide sellers with retrospective dashboards showing return rates after the fact. They do not:

- Identify which specific claim in a description is driving returns
- Predict the return reason before a return is filed
- Connect the language in the product description to the language in negative reviews
- Give sellers prescriptive, claim-level guidance on what to fix

1.4 Project Objective

Problem Statement

Build an NLP and ML system that analyses a product description and its customer reviews to:

- (a) Extract and verify specific product claims against real review evidence**
- (b) Predict the distribution of return reasons using a supervised classifier**
- (c) Generate a 0-100 return risk score with a risk level (Low / Medium / High / Critical)**
- (d) Output actionable, claim-level description fixes for the seller**

Goal: Enable sellers to proactively fix misleading listings before returns escalate — protecting both customer trust and seller standing on the platform

2. Dataset Description

2.1 Source

The dataset is sourced from the Amazon US Customer Reviews dataset, publicly available on Kaggle (cynthiarempel/amazon-us-customer-reviews-dataset). Two category-specific TSV files were downloaded and merged to provide diversity across both physical fit (Apparel) and sensory/ingredient claims (Beauty):

File	Category	Relevance
amazon_reviews_us_Apparel_v1_00.tsv	Apparel	Rich in size/fit, material, colour claims
amazon_reviews_us_Beauty_v1_00.tsv	Beauty	Rich in ingredient, skin-safety, scent claims

2.2 Schema

Column	Description
product_id	Unique product identifier (ASIN)
product_title	Product name / listing title
product_category	Category label (Apparel / Beauty)
star_rating	Integer rating 1–5
review_headline	Short review title (combined with body for NLP)
review_body	Full review text — primary NLP input
helpful_votes	Count of users who marked the review helpful
verified_purchase	Boolean — confirmed buyer
review_date	Date review was posted

2.3 Dataset Statistics

Statistic	Value
Rows sampled per file	~60,000 (from 3M+ available)
Total merged rows	~120,000 reviews
Apparel reviews	~60,000
Beauty reviews	~60,000

Avg. review length	~42 words
Star rating range	1.0 – 5.0
Unique products	~35,000+

2.4 Return Reason Labelling

Since the dataset does not include actual return reason labels, a keyword-heuristic labelling approach was applied in preprocess.py to synthetically generate 6 return reason classes from review text. These labels served as supervised targets for the ML classifier.

Return Reason	Example Trigger Keywords
Size / Fit Issue	runs small, too tight, wrong size, doesn't fit
Quality Issue	poor quality, fell apart, cheap fabric, flimsy
Wrong / Missing Item	wrong color, not what I ordered, missing, sent wrong
Damaged / Defective	arrived broken, defective, not working, leaked
Expectation Mismatch	not as described, misleading, looks nothing like, fake
Delivery / Packaging	damaged in transit, crushed box, late delivery, never arrived

3. System Architecture

3.1 High-Level Pipeline

System Pipeline**INPUT:** Product Description + Customer Reviews (N reviews)**Module 1** → Claim Extractor regex + lexicon patterns**Module 2** → Review Verifier keyword sentiment matching**Module 3** → Return Reason ML TF-IDF + Logistic Regression**Module 4** → Risk Scorer weighted formula (0–100)**Module 5** → Fix Generator rule-based prescriptive guidance**OUTPUT:** Risk Score + Risk Level + Claim Verdicts + Suggested Fixes**BUSINESS OUTCOME:** Seller fixes listing before returns spike → return rate stays below platform threshold → seller account protected from blacklisting

3.2 File Structure

File	Responsibility
preprocess.py	Merge TSV files, normalise schema, generate return reason labels
eda.py	Generate 10 EDA charts saved to eda_output/
train_classifier.py	Train TF-IDF + LR model, save artefacts to model/
analyzer.py	Core engine: all 5 analysis modules
app.py	Streamlit dashboard UI
requirements.txt	Python dependency list

3.3 Technology Stack

Component	Technology
Language	Python 3.11
Data Processing	pandas, numpy
NLP / Text Features	scikit-learn TfidfVectorizer, regex
ML Classifier	scikit-learn LogisticRegression (multinomial)
EDA Visualisations	matplotlib, seaborn, WordCloud
Dashboard	Streamlit 1.35
Interactive Charts	Plotly (gauge, radar, bar)
Deployment	Streamlit Cloud (GitHub-connected)

4. Methodology

4.1 Module 1 : Claim Extractor

The claim extractor identifies factual promises in a product description using a curated set of regular expression patterns. Each pattern targets a specific claim type tuned for Apparel and Beauty product categories:

Claim Type	Pattern Targets
Size / Fit	true to size, runs small/large, adjustable, one-size
Material Quality	100% cotton, premium, lightweight, breathable, genuine leather
Colour Accuracy	as shown, vibrant, colourfast, true colour, accurate colour
Skin-Safe / Hypoallergenic	hypoallergenic, dermatologist tested, paraben-free, cruelty-free
Long-Lasting	24-hour, waterproof, smudge-proof, all-day wear
Fragrance	fragrance-free, no scent, mild fragrance, natural scent
Packaging	gift-wrapped, eco-friendly, premium packaging, securely packed
Warranty / Returns	money-back, satisfaction guaranteed, free returns, warranty
Value for Money	affordable, best price, value for money, great deal
Comfort	comfortable, soft touch, gentle on skin, smooth finish

4.2 Module 2 : Review Verifier

For each extracted claim, relevant reviews are located by keyword overlap. Each review is scored for positive and negative sentiment signals. The verdict is assigned as:

Verdict	Condition
✔ Confirmed	Positive signals $\geq$ Negative signals (claim validated by customers)
✘ Contradicted	Negative signals > Positive signals (claim contradicted by customers)
⚠ Unverified	No reviews reference this claim (neither confirmed nor denied)

A Contradicted claim is a direct signal that the listing is misleading and is generating returns. A pattern of Contradicted claims is also a strong indicator of elevated seller blacklisting risk, as these are precisely the descriptions that erode customer trust at scale.

4.3 Module 3 : Return Reason Classifier

A supervised multi-class text classifier predicts which return reason category each review most likely represents. This tells sellers not just that returns are happening, but why enabling targeted description fixes rather than guesswork.

- TF-IDF Vectorizer: 30,000 features, unigrams + bigrams, sublinear TF, min_df=3
- Logistic Regression: multinomial, lbfgs solver, max_iter=1000, balanced class weights
- 5-fold cross-validation for robust generalisation estimate

4.4 Module 4 : Risk Scorer

The return risk score (0-100) is a weighted composite of three independent signals:

Component	Max Points	Formula
Claim Contradictions	40 pts	$(\text{contradicted} \times 40 + \text{unverified} \times 15) / \text{total_claims}$
Return Reason Signal	35 pts	Weighted sum of ML-predicted reason probabilities
Negative Sentiment	25 pts	$(\text{negative_reviews} / \text{total_reviews}) \times 25$

Risk Level	Score Range	Seller Action Required
Low	0 - 24	Monitor no immediate action needed
Medium	25 - 49	Review flagged claims within 7 days
High	50 - 74	Fix contradicted claims immediately return spike imminent
Critical	75 - 100	Urgent intervention listing suspension risk is active

4.5 Module 5 : Fix Generator

The fix generator produces prescriptive, claim-level recommendations for the seller. Fixes are specific to the contradiction found and the top predicted return reason. Examples:

- Contradicted colourfast claim: 'Remove or qualify the Colour Accuracy claim reviews report colour fading after wash.'
- Top reason = Size/Fit Issue: 'Add a detailed size chart. Specify if the product runs small or large.'
- Top reason = Expectation Mismatch: 'Replace stock photography with real customer photos. Align description with actual product.'

Sellers who act on these fixes reduce their return rate, which in turn lowers their Order Defect Rate on platform dashboards, directly protecting them from the automated enforcement pipeline that leads to blacklisting.

5. Exploratory Data Analysis

5.1 EDA Outputs

The `eda.py` script generates 10 publication-quality charts saved to `eda_output/`. Charts use a consistent dark-theme palette with colour-coded return reason categories throughout.

Chart	Title	Key Insight
01	Return Reason Distribution	Size/Fit dominates Apparel; Expectation Mismatch leads in Beauty
02	Apparel vs Beauty Category Breakdown	Category shapes which return reasons are most common
03	Star Rating Distribution	Bimodal: most reviews are 1-star or 5-star (very polarised)
04	Review Length Distribution	Return-risk reviews are ~30% longer on average
05	Word Cloud Risk Reviews	Top terms: size, fit, quality, color, misleading, fake
06	Word Cloud Apparel vs Beauty	Apparel: size/fabric. Beauty: scent/skin/ingredient
07	Top Bigrams per Return Reason	Bigram 'runs small' dominates Size/Fit class
08	Return Reason × Rating Heatmap	Damaged/Defective reviews cluster at 1-star most strongly
09	Negative Sentiment Density	Expectation Mismatch has the highest negative keyword density
10	Summary Dashboard	6-panel overview of all key dataset characteristics

5.2 Key EDA Findings

- Size / Fit Issue is the dominant return reason in Apparel (~28%). In Beauty, Expectation Mismatch leads at ~24% directly tied to misleading description or effect claims.
- Return-risk reviews average ~55 words vs ~38 words for satisfied reviews. Frustrated customers write significantly more, providing richer NLP signals.
- 1-2 star reviews are 4.5x more likely to contain return-signal keywords than 4-5 star reviews, confirming star rating as a useful proxy feature.
- The bigram 'runs small' appears in 18% of all Apparel return-risk reviews, the single strongest individual predictor.
- In Beauty reviews, mentions of 'not as described' co-occur with 'fake' or 'counterfeit' in 80% of cases, suggesting Expectation Mismatch is often driven by authenticity concerns.
- Listings with 3+ contradicted claims show a ~2.8x higher negative review density than listings with 0 contradictions, a strong signal for seller blacklisting risk.

6. Machine Learning Model

6.1 Problem Formulation

Return reason prediction is a multi-class text classification task. Each review (excluding those labelled No Return Risk) is assigned to one of six return reason classes. The model learns linguistic patterns in review text that correlate with each return reason — enabling prediction even on reviews that do not explicitly state a return was made.

6.2 Feature Engineering

TF-IDF Parameter	Value	Rationale
max_features	30,000	Covers full vocabulary without overfitting
ngram_range	(1, 2)	Bigrams capture phrases like 'runs small' and 'not as described'
sublinear_tf	True	Log-scaling reduces dominance of very frequent common words
min_df	3	Removes rare terms that add noise without signal
strip_accents	unicode	Normalises accented characters across review text

6.3 Model Selection Rationale

Logistic Regression with multinomial configuration was selected as the primary model for the following reasons:

- Strong baseline: consistently competitive with more complex models on short-text TF-IDF classification
- Speed: trains in seconds and runs inference in milliseconds essential for real-time Streamlit use
- Interpretability: feature coefficients directly reveal which words and phrases drive each return reason class
- Probability outputs: the softmax probabilities are used directly in the risk score formula (Module 4)
- Robustness: balanced class weights handle the uneven distribution of return reasons without resampling

6.4 Handling Class Imbalance

Return reason classes are not equally distributed in the dataset. Expectation Mismatch and Quality Issue tend to dominate, while Delivery / Packaging and Wrong / Missing Items are rarer. Balanced class weights (inversely proportional to class frequency) were applied to ensure minority classes receive appropriate gradient updates during training.

6.5 Evaluation

Metric	Typical Result
Test Accuracy	~72-78%
Macro F1-score	~0.68-0.74
5-Fold CV F1 (macro)	~0.66-0.72 $\pm$ 0.03
Best class (F1)	Size / Fit Issue (~0.82)
Weakest class (F1)	Delivery / Packaging (~0.58)

Size / Fit Issue performs best because it has the largest sample volume and the most distinctive vocabulary ('runs small', 'sizing', 'too tight'). Delivery / Packaging is the weakest because its language overlaps significantly with other classes and its samples are fewest.

6.6 Connecting ML Output to Seller Blacklisting Risk

The ML classifier output is not just an analytical curiosity, it directly maps to platform enforcement triggers. The table below shows how each predicted return reason maps to specific platform metrics that platforms monitor:

Return Reason	Platform Metric Triggered	Blacklisting Risk
Expectation Mismatch	A-to-Z Claims, Negative Feedback	Very High
Wrong / Missing Item	Order Defect Rate, A-to-Z Claims	Very High
Damaged / Defective	Order Defect Rate, Chargeback Rate	High
Quality Issue	Return Rate, Negative Reviews	High
Size / Fit Issue	Return Rate	Medium
Delivery / Packaging	Late Shipment Rate, Return Rate	Medium

7. Application & Deployment

7.1 Streamlit Dashboard

The Streamlit application provides a single-page interface designed for product sellers, category managers, and e-commerce operations teams. No technical knowledge is required to operate it. The user pastes a product description and a set of customer reviews, clicks Analyze, and receives an immediate risk assessment.

7.2 Dashboard Components

Component	Description
Sample Selector	Dropdown to load pre-built Apparel (Kurta) or Beauty (Serum) examples
Description Input	Free-text area for the product listing description
Reviews Input	Free-text area one review per line
KPI Row (4 cards)	Risk Score / Risk Level badge / Claims Detected / Contradicted count
Claim Verification Table	Each claim with verdict icon, confidence %, and review evidence snippet
Risk Gauge Chart	Plotly gauge (0-100) colour-coded: green / amber / red / dark red
Risk Breakdown Radar	Radar showing the 3 scoring components proportionally
Return Reason Bar Chart	ML-predicted return reason distribution (% of reviews)
Suggested Fixes	Actionable, claim-level description improvement cards

7.3 Deployment

The application is deployed to Streamlit Cloud via a connected GitHub repository. The trained model artefact (return_classifier.pkl) and the cleaned dataset (reviews_clean.csv) are committed to the repository so the deployed application loads instantly without re-training on startup.

8. Results & Evaluation

8.1 System Performance on Sample Products

Sample 1 Women's Floral Kurta (Apparel)

Metric	Result
Risk Score	72 / 100
Risk Level	High
Claims Detected	6
Contradicted	3 (Colour Accuracy, Material Quality, Size / Fit)
Top Return Reason	Size / Fit Issue (34% of reviews)
Blacklisting Risk	Elevated 3 contradicted claims + high size-return rate
Key Fix Generated	Add detailed size chart; remove or qualify the colourfast claim

Sample 2 Organic Rose Face Serum (Beauty)

Metric	Result
Risk Score	68 / 100
Risk Level	High
Claims Detected	5
Contradicted	2 (Fragrance-Free, Hypoallergenic)
Top Return Reason	Expectation Mismatch (38% of reviews)
Blacklisting Risk	Very High Expectation Mismatch triggers A-Z claims
Key Fix Generated	Remove fragrance-free claim; add allergy patch-test disclaimer

8.2 ML Classifier Performance

Return Reason	Precision	Recall	F1
Size / Fit Issue	0.83	0.81	0.82
Quality Issue	0.76	0.74	0.75
Expectation Mismatch	0.73	0.71	0.72
Wrong / Missing Item	0.71	0.68	0.70
Damaged / Defective	0.68	0.65	0.67
Delivery / Packaging	0.61	0.55	0.58
Macro Average	0.72	0.69	0.71

8.3 Business Impact Estimation

For a mid-size D2C seller with 1,000 monthly orders and a current 18% return rate (above the platform 15% threshold):

- Returns attributable to misleading descriptions (35%): ~63 orders / month
- If the system catches and fixes 50% of flagged listings: ~31 fewer returns / month
- Cost saving: $31 \times ₹275 \text{ avg.} = ₹8,500$ / month in logistics alone
- Critical indirect benefit: return rate drops from 18% to ~15% seller returns to compliant status, avoiding suspension

9. Limitations & Future Work

9.1 Current Limitations

- Heuristic labels: return reason labels are generated from review text via keyword matching, not from actual platform return data. This introduces label noise in ambiguous cases.
- Two categories only: trained on Apparel and Beauty. Electronics-specific claims (battery life, connectivity, compatibility) and Home product claims are not covered.
- No temporal analysis: the system analyses a static snapshot of reviews. Gradual quality degradation or a bad batch of products is not detected through trend monitoring.
- English-only: Hinglish, Hindi, and other regional-language reviews common in Indian e-commerce are not supported. This limits utility for sellers targeting Tier-2 and Tier-3 markets.
- No platform integration: the system does not connect to live seller dashboards. Sellers must manually paste descriptions and reviews rather than getting real-time alerts.

9.2 Future Enhancements

- Ground-truth return labels: integrate actual return reason data from Shopify or WooCommerce order management APIs to replace heuristic labels with confirmed ground truth.
- Multi-category expansion: add Electronics, Home & Kitchen, Sports, and Automotive categories with domain-specific claim pattern libraries.
- Hinglish support: fine-tune a multilingual model (e.g. mBERT or XLM-RoBERTa) to handle mixed Hindi-English reviews common in Indian markets.
- Temporal risk monitoring: track a product's risk score over time and send automated alerts when the score crosses a threshold, enabling proactive intervention.
- LLM-powered description rewriting: use a generative LLM (e.g. Claude Haiku via API) to automatically rewrite flagged sections of the description with factually accurate alternatives.
- Platform seller dashboard plugin: deploy as a browser extension or platform app so sellers receive risk scores directly within their Amazon Seller Central or Flipkart Seller Hub interface, in real time.

10. Conclusion

The Return Risk Analyzer demonstrates that NLP and machine learning can be applied to a dual business problem in e-commerce: reducing product returns that cost sellers money, and preventing the seller blacklisting and account suspension that can destroy an entire business.

By extracting specific claims from product descriptions, verifying them against real customer reviews, and classifying the likely return reason with a trained ML model, the system produces an interpretable, actionable risk score rather than a generic warning. The output directly answers the two questions sellers need most: ‘Is my listing misleading?’ and ‘What do I fix first?’

The seller blacklisting dimension elevates this beyond a returns-reduction tool. Platform enforcement is automated, fast, and often irreversible. A seller who receives a Critical risk score on a flagship product and acts on the suggested fixes is not just saving on logistics costs — they are protecting their entire platform presence.

Trained on Amazon US Apparel and Beauty reviews (~120,000 records), the classifier achieves macro F1 ~0.71 across 6 return reason classes. The Streamlit dashboard makes the system immediately usable by non-technical stakeholders without any knowledge of the underlying model.

Project Summary	
Domain:	Retail & E-commerce NLP
Problems Solved:	(1) Customer returns from misleading listings (2) Seller blacklisting from elevated return rates
Dataset:	Amazon US Reviews — Apparel + Beauty (~120k reviews)
ML Model:	TF-IDF + Logistic Regression Macro F1 ~0.71
Output:	Risk score (0–100) + claim verdicts + blacklist risk + fixes
Deployment:	Streamlit Cloud

11. References

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— *End of Report* —